SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

BHARAT BALLOT

Secure Digital Election Management Platform

Version: 1.0
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Project: Bharat Ballot
Document Type: Software Requirements Specification (SRS)

Chapter 1 — Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the functional and non-functional requirements of **Bharat Ballot**, a secure, AI-powered digital election management platform designed to modernize the electoral process through advanced software engineering, biometric identity verification, and role-based administrative controls.

This document serves as the primary reference for the design, development, testing, deployment, and future enhancement of the Bharat Ballot platform. It establishes a common understanding of the system's objectives, architecture, capabilities, and operational requirements among developers, project stakeholders, evaluators, and future contributors.

The SRS also provides a structured description of the platform's features, system behavior, security mechanisms, user roles, and technical constraints, ensuring that every component of the application aligns with the project's vision of delivering secure, transparent, and trustworthy digital elections.

1.2 Scope

Bharat Ballot is a comprehensive web-based election management platform that digitizes the complete election lifecycle while maintaining the integrity and transparency expected from democratic institutions.

The platform supports multiple categories of elections, including:

- Lok Sabha Elections
- State Assembly Elections
- Municipal Corporation Elections
- Panchayat Elections

The system provides dedicated portals for election authorities and citizens, enabling secure management of elections, candidates, constituencies, voters, and voting operations through a centralized architecture.

Key capabilities of the platform include:

- Election lifecycle management
- Candidate registration and administration
- Constituency-based administrative control
- Secure voter registration
- AI-powered facial enrollment and verification
- One-person, one-vote enforcement
- Secure vote casting
- Automated result publication
- Role-based authentication and authorization
- Cloud-based image management
- RESTful service communication

Unlike conventional online voting systems that primarily digitize existing manual processes, Bharat Ballot incorporates artificial intelligence, biometric authentication, and modular service architecture to improve election security, administrative efficiency, and public trust.

1.3 Intended Audience

This document is intended for the following stakeholders:

- Software Developers
- System Architects
- Project Evaluators
- Academic Supervisors
- Testing Engineers
- Future Contributors
- System Administrators
- Election Management Personnel

It serves as both a technical reference during development and a formal specification for project evaluation and maintenance.

1.4 Objectives

The primary objectives of Bharat Ballot are:

- To provide a secure and transparent digital election platform.
- To eliminate unauthorized or duplicate voting through biometric verification.
- To simplify election administration using role-based management.
- To automate election scheduling and lifecycle management.
- To ensure constituency-specific administrative control.
- To provide real-time and reliable election result publication.
- To leverage artificial intelligence for voter identity verification.
- To create a scalable architecture capable of supporting multiple election types.
- To improve accessibility while maintaining election integrity.
- To establish a modern, cloud-enabled election management ecosystem.

1.5 Definitions, Acronyms and Abbreviations

Term	Description
AI	Artificial Intelligence
API	Application Programming Interface
JWT	JSON Web Token
REST	Representational State Transfer
SRS	Software Requirements Specification
UI	User Interface
UX	User Experience
ODM	Object Document Mapper
RBAC	Role-Based Access Control
EPIC	Elector's Photo Identity Card

Term	Description
Face Embedding	Numerical representation of facial features generated by the AI service
InsightFace	Deep learning framework used for face recognition
Cosine Similarity	Mathematical technique used to compare face embeddings
Command Center	Administrative portal used by Heads and Administrators
Citizen Portal	Portal used by registered voters

1.6 Document Overview

This Software Requirements Specification is organized into the following chapters:

Chapter	Description
Chapter 1	Introduction
Chapter 2	Overall Description
Chapter 3	System Features and Functional Requirements
Chapter 4	External Interface Requirements
Chapter 5	Non-Functional Requirements
Chapter 6	System Models and Diagrams
Chapter 7	Future Scope
Chapter 8	Appendices and References

Chapter Summary

This introductory chapter establishes the foundation of the Bharat Ballot Software Requirements Specification by defining the project's purpose, scope, objectives, stakeholders, terminology, and overall document structure. The subsequent chapters build upon this foundation to describe the system's architecture, functional modules, security mechanisms, and operational requirements in detail.

2.1 Product Perspective

Bharat Ballot is a next-generation digital election management platform developed to modernize the electoral process through secure web technologies, artificial intelligence, and cloud-based infrastructure. The platform serves as a complete election ecosystem capable of managing the entire election lifecycle, from election creation and voter registration to secure vote casting and result publication.

Unlike traditional election management systems that primarily digitize manual workflows, Bharat Ballot integrates biometric identity verification, role-based administrative control, and automated election lifecycle management into a unified platform. The system follows a modular, service-oriented architecture consisting of independent frontend applications, a centralized RESTful backend, a dedicated AI-powered face recognition service, and cloud-based storage.

The platform is designed to support multiple categories of elections, including Lok Sabha, State Assembly, Municipal Corporation, and Panchayat elections. Through dedicated interfaces for Heads, Administrators, and Citizens, Bharat Ballot provides secure, scalable, and transparent election management while maintaining the integrity expected from democratic institutions.

2.2 Product Functions

Bharat Ballot provides a comprehensive set of functions required for managing digital elections securely and efficiently. The primary system functions include:

Election Management

- Create and configure elections.
- Define election type, schedule, and duration.
- Manage election lifecycle (Draft, Active, Completed, Archived).
- Publish election results.

User Management

- Manage Head accounts.
- Manage Administrator accounts.
- Register and manage voters.
- Assign administrators to authorized constituencies.

Candidate Management

- Register election candidates.
- Update candidate information.

- Manage party affiliation and candidate status.
- Associate candidates with specific constituencies and elections.

Constituency Management

- Maintain master constituency records.
- Generate election-specific constituencies.
- Support Lok Sabha, Assembly, Municipal, and Panchayat elections.
- Map administrative jurisdictions.

Secure Voting

- Authenticate voters.
- Perform biometric face verification.
- Display eligible candidates.
- Record secure digital votes.
- Prevent duplicate voting.

Result Management

- Calculate election results.
- Display constituency-wise outcomes.
- Publish official election results.
- Maintain election records.

AI Face Recognition

- Enroll voter facial embeddings.
- Verify voter identity before voting.
- Compare facial embeddings using cosine similarity.
- Ensure secure biometric authentication.

2.3 User Classes and Characteristics

The platform supports three primary categories of users.

Head (Election Commissioner)

The Head acts as the highest authority within the platform and possesses system-wide administrative privileges.

Responsibilities include:

- Managing elections.
- Managing administrators.
- Monitoring election activities.
- Publishing results.
- Supervising overall election operations.

Characteristics:

- Advanced system privileges.
- Access to all election data.
- Responsible for governance and policy enforcement.

Administrator

Administrators operate within assigned constituencies and perform constituency-level election management.

Responsibilities include:

- Managing voters.
- Managing candidates.
- Monitoring constituency elections.
- Maintaining voter information.
- Verifying election records.

Characteristics:

- Restricted to assigned constituencies.
- Role-based access permissions.
- Responsible for operational election management.

Citizen (Voter)

Citizens are registered voters authorized to participate in elections.

Responsibilities include:

- Logging into the Citizen Portal.
- Completing face verification.
- Viewing election candidates.

- Casting a secure vote.
- Viewing election results.

Characteristics:

- Limited access privileges.
- One-person, one-vote restriction.
- Biometric authentication before voting.

2.4 Operating Environment

Bharat Ballot operates within a modern web-based environment utilizing cloud infrastructure and distributed services.

Client Environment

- Modern web browsers (Chrome, Edge, Firefox, Safari)
- Desktop and laptop computers
- Responsive interface for various screen sizes

Server Environment

- Node.js runtime
- Express.js REST API
- MongoDB Atlas
- Cloudinary cloud storage

AI Environment

- Python 3.10
- FastAPI
- InsightFace
- OpenCV
- ONNX Runtime

Deployment Environment

- Render (Frontend & Backend Hosting)
- MongoDB Atlas
- Cloudinary

2.5 Design and Implementation Constraints

The system has been developed while considering the following constraints:

- Internet connectivity is required for all platform operations.
- Face verification depends on webcam availability and image quality.
- AI verification accuracy depends on facial visibility and lighting conditions.
- Only authorized administrators may manage assigned constituencies.
- Election operations are restricted according to election status.
- Secure authentication is mandatory before accessing protected resources.
- Cloud storage services are required for image management.
- MongoDB Atlas availability directly affects system accessibility.

2.6 User Documentation

The following documentation accompanies the Bharat Ballot platform:

- Software Requirements Specification (SRS)
- Software Design Document (SDD)
- User Manual
- Administrator Manual
- API Documentation
- Database Documentation
- Deployment Guide
- Security Documentation

2.7 Assumptions and Dependencies

The successful operation of Bharat Ballot depends upon the following assumptions:

- Users possess stable internet connectivity.
- Administrators maintain accurate election and voter records.
- MongoDB Atlas remains operational.
- Cloudinary services are available for image storage.
- The AI face recognition service is continuously accessible.

- Users provide valid identification during voter registration.
- Supported web browsers maintain JavaScript compatibility.
- Server and client system clocks remain synchronized for election scheduling.

Chapter Summary

This chapter presented the overall description of the Bharat Ballot platform, including its purpose within the digital election ecosystem, major system functions, user classifications, operational environment, design constraints, supporting documentation, and external dependencies. These descriptions provide the contextual foundation for the detailed functional and non-functional requirements discussed in the following chapters.

Chapter 3 — Functional Requirements

3.1 Introduction

This chapter defines the functional requirements of the Bharat Ballot platform. Functional requirements describe the specific services, operations, and behaviors that the system must provide to users. Each module of the platform is designed to perform a well-defined set of responsibilities while ensuring security, transparency, scalability, and reliability throughout the election lifecycle.

The Bharat Ballot platform follows a modular architecture in which each subsystem performs independent functions while communicating through secure REST APIs. This modular design improves maintainability, scalability, and system reliability.

The major functional modules of the system include:

- Authentication Module
- Election Management Module
- Administrator Management Module
- Candidate Management Module
- Voter Management Module
- Constituency Management Module
- AI Face Recognition Module
- Voting Module
- Result Management Module

3.2 Authentication Module

Purpose

The Authentication Module is responsible for securely verifying user identities before granting access to protected resources. It ensures that only authorized users can access system functionalities according to their assigned roles.

The module supports authentication for Heads, Administrators, and Citizens while implementing secure password hashing, JWT-based authentication, and role-based authorization.

Inputs

- Username / EPIC Number

- Password
- User Role

Processing

The system performs the following operations:

- Validate user credentials.
- Verify password using bcrypt.
- Generate JWT authentication token.
- Assign user role.
- Redirect user to the appropriate dashboard.
- Reject unauthorized requests.

Outputs

- Successful authentication
- JWT Token
- User session
- Authorization status

Functional Requirements

FR-1.1

The system shall authenticate Heads using registered credentials.

FR-1.2

The system shall authenticate Administrators using registered credentials.

FR-1.3

The system shall authenticate Citizens using their EPIC number and password.

FR-1.4

The system shall encrypt passwords before storing them.

FR-1.5

The system shall generate JWT tokens after successful authentication.

FR-1.6

The system shall prevent unauthorized access to protected routes.

FR-1.7

The system shall invalidate expired authentication tokens.

FR-1.8

The system shall enforce role-based authorization.

3.3 Election Management Module

Purpose

The Election Management Module enables the creation, configuration, monitoring, and lifecycle management of elections. It acts as the central component governing all election-related operations within the platform.

The module supports multiple election types and automatically manages election status transitions based on configured schedules.

Inputs

- Election Name
- Election Type
- Election Schedule
- Constituencies
- Result Visibility
- Election Status

Processing

The module performs:

- Election creation.
- Election scheduling.
- Election activation.
- Election completion.
- Election archival.

- Constituency generation.
- Result publication.

Outputs

- Election records
- Election status
- Generated constituencies
- Election dashboard

Functional Requirements

FR-2.1

The system shall allow the Head to create elections.

FR-2.2

The system shall support Lok Sabha elections.

FR-2.3

The system shall support State Assembly elections.

FR-2.4

The system shall support Municipal elections.

FR-2.5

The system shall support Panchayat elections.

FR-2.6

The system shall automatically synchronize election status.

FR-2.7

The system shall prevent modifications after election completion.

FR-2.8

The system shall archive completed elections.

3.4 Administrator Management Module

Purpose

This module manages constituency-level election administrators responsible for conducting election activities within their assigned jurisdictions.

Each Administrator operates only within authorized constituencies using role-based permissions.

Functional Requirements

FR-3.1

The system shall allow the Head to create Administrator accounts.

FR-3.2

The system shall assign Administrators to specific constituencies.

FR-3.3

The system shall restrict Administrators to assigned constituencies.

FR-3.4

The system shall allow updating Administrator information.

FR-3.5

The system shall allow removal of Administrators.

3.5 Candidate Management Module

Purpose

The Candidate Management Module enables election authorities to register, manage, and maintain candidate information for different election types.

Functional Requirements

FR-4.1

The system shall register candidates.

FR-4.2

The system shall associate candidates with elections.

FR-4.3

The system shall associate candidates with constituencies.

FR-4.4

The system shall update candidate information.

FR-4.5

The system shall delete candidate records.

FR-4.6

The system shall prevent duplicate candidate registrations.

3.6 Voter Management Module

Purpose

This module manages voter registration, verification, profile maintenance, and eligibility validation.

Functional Requirements

FR-5.1

The system shall register voters.

FR-5.2

The system shall store voter photographs securely.

FR-5.3

The system shall generate facial embeddings during registration.

FR-5.4

The system shall allow profile updates.

FR-5.5

The system shall verify voter eligibility.

FR-5.6

The system shall maintain voting status.

3.7 Constituency Management Module

Purpose

This module manages election constituencies and geographical divisions for different election categories.

Functional Requirements

FR-6.1

The system shall maintain master constituency records.

FR-6.2

The system shall generate election-specific constituencies.

FR-6.3

The system shall support constituency assignment.

FR-6.4

The system shall prevent invalid constituency mapping.

3.8 AI Face Recognition Module

Purpose

The AI Face Recognition Module provides biometric identity verification using InsightFace before allowing a citizen to cast a vote.

Functional Requirements

FR-7.1

The system shall extract facial embeddings during registration.

FR-7.2

The system shall securely store facial embeddings.

FR-7.3

The system shall capture a live facial image before voting.

FR-7.4

The system shall compare live and stored embeddings.

FR-7.5

The system shall deny voting when identity verification fails.

FR-7.6

The system shall allow voting only after successful biometric verification.

3.9 Voting Module

Purpose

The Voting Module enables eligible citizens to cast secure digital votes after successful authentication and biometric verification.

Functional Requirements

FR-8.1

The system shall display eligible candidates.

FR-8.2

The system shall allow only one vote per voter.

FR-8.3

The system shall prevent duplicate voting.

FR-8.4

The system shall securely record votes.

FR-8.5

The system shall update voting status immediately after vote submission.

3.10 Result Management Module

Purpose

The Result Management Module is responsible for calculating, publishing, and presenting election results.

Functional Requirements

FR-9.1

The system shall calculate vote counts.

FR-9.2

The system shall determine winning candidates.

FR-9.3

The system shall publish results after election completion.

FR-9.4

The system shall display constituency-wise results.

FR-9.5

The system shall preserve election history.

Chapter Summary

This chapter defined the functional requirements of Bharat Ballot by describing each core module and its associated responsibilities. The functional requirements specify how the platform manages authentication, elections, administrators, candidates, voters, constituencies, biometric verification, voting, and result publication. Together, these modules form the operational foundation of the Bharat Ballot system and ensure secure, transparent, and efficient digital election management.

Chapter 4 — External Interface Requirements

4.1 Introduction

The External Interface Requirements define the interactions between the Bharat Ballot platform and external entities, including users, hardware devices, third-party services, and software components. These interfaces enable seamless communication between different modules while maintaining security, reliability, and scalability.

Bharat Ballot adopts a service-oriented architecture where independent frontend applications communicate with a centralized backend through RESTful APIs. The backend further interacts with cloud storage services, database systems, and an AI-powered biometric verification service.

The external interfaces are categorized into:

- User Interfaces
- Hardware Interfaces
- Software Interfaces
- Communication Interfaces

4.2 User Interface Requirements

The Bharat Ballot platform provides dedicated user interfaces tailored to different categories of users. Each interface is designed to present only the functionalities relevant to the authenticated user's role.

4.2.1 Citizen Portal

The Citizen Portal enables registered voters to securely participate in elections.

Major interface components include:

- Home Page
- Voter Login
- Dashboard
- Election Selection
- Candidate List
- Face Verification Screen
- Voting Interface
- Election Results
- Profile Management

Interface Characteristics

- Responsive design
- Clean and intuitive navigation
- Secure authentication
- Real-time election status
- Accessibility-focused layout

4.2.2 Command Center

The Command Center serves as the administrative portal for Heads and Administrators.

Major interface components include:

Head Dashboard

- Election Management
- Administrator Management
- Dashboard Analytics
- Result Publication
- Election Monitoring

Administrator Dashboard

- Candidate Management
- Voter Registration
- Constituency Management
- Election Monitoring
- Result Viewing

Interface Characteristics

- Role-based navigation
- Administrative dashboards
- Data tables
- Interactive forms
- Search and filtering capabilities
- Responsive layouts

4.3 Hardware Interface Requirements

The Bharat Ballot platform interacts with standard computer hardware and peripheral devices.

Supported Hardware

Client Devices

- Desktop Computers
- Laptop Computers
- Modern Web Browsers

Camera

The platform requires webcam access for biometric verification during the voting process.

Camera requirements include:

- Functional webcam
- Adequate lighting conditions
- Face visibility
- Browser permission for camera access

Server Hardware

The backend services are hosted on cloud infrastructure and do not require specialized hardware on the client side.

4.4 Software Interface Requirements

The Bharat Ballot platform communicates with several software components and third-party services.

React Frontend

Purpose:

Provides interactive user interfaces for Citizens, Administrators, and Heads.

Responsibilities:

- Authentication
- Dashboard rendering
- API communication

- Form validation
- Navigation
- User interaction

Express.js Backend

Purpose:

Acts as the centralized REST API server.

Responsibilities:

- Authentication
- Business logic
- Database communication
- Election management
- Candidate management
- Voter management
- Result generation
- AI service communication

MongoDB Atlas

Purpose:

Stores all persistent platform data.

Stored information includes:

- Elections
- Candidates
- Voters
- Administrators
- Constituencies
- Votes
- Authentication records

AI Face Recognition Service

Purpose:

Provides biometric identity verification.

Responsibilities:

- Face detection
- Face embedding generation
- Face enrollment
- Face verification
- Similarity comparison

The AI service communicates with the backend through secure REST APIs.

Cloudinary

Purpose:

Cloud-based image storage.

Responsibilities:

- Store voter photographs
- Store candidate photographs
- Secure image retrieval
- Image optimization

4.5 Communication Interface Requirements

Communication between system components is performed using secure HTTP-based RESTful APIs.

Client to Backend Communication

Protocol:

- HTTPS

Data Format:

- JSON

Authentication:

- JWT Tokens

Communication includes:

- Login requests
- Election requests
- Candidate requests
- Voting requests
- Dashboard requests

Backend to Database Communication

Database:

MongoDB Atlas

Communication Method:

Mongoose ODM

Purpose:

- Store records
- Retrieve records
- Update records
- Delete records

Backend to AI Service Communication

Protocol:

HTTP REST API

Purpose:

- Face enrollment
- Face verification

Data exchanged includes:

- Uploaded facial image
- Generated embeddings
- Similarity scores
- Verification results

Backend to Cloudinary Communication

Purpose:

Image upload and retrieval.

Communication includes:

- Image upload
- Image deletion
- Secure URL generation

4.6 Interface Constraints

The following constraints apply to system interfaces:

- Internet connectivity is mandatory.
- Browser camera permission is required for face verification.
- All protected APIs require valid JWT authentication.
- Only authorized users may access restricted interfaces.
- Communication between services must follow predefined REST API specifications.
- JSON is used as the standard data exchange format.
- External cloud services must remain available for image storage and retrieval.

4.7 Error Handling and User Feedback

The system provides informative feedback whenever an interface operation cannot be completed successfully.

Examples include:

- Invalid login credentials
- Camera permission denied
- Face verification failed
- Election unavailable
- Unauthorized access
- Network communication failure

- Internal server error

Appropriate success, warning, and error messages are displayed to guide users while preventing disclosure of sensitive system information.

Chapter Summary

This chapter described the external interfaces of the Bharat Ballot platform, including user interfaces, hardware requirements, software integrations, and communication mechanisms. It established how different system components interact securely through standardized protocols while maintaining reliable communication between the frontend applications, backend services, cloud infrastructure, and AI-powered biometric verification system.

Chapter 5 — Non-Functional Requirements

5.1 Introduction

Non-functional requirements define the quality attributes, operational standards, and performance expectations of the Bharat Ballot platform. Unlike functional requirements, which specify what the system should do, non-functional requirements describe how the system should perform under various operating conditions.

As a digital election management platform, Bharat Ballot emphasizes security, reliability, scalability, performance, usability, and maintainability to ensure a trustworthy and efficient election process. These requirements guide the design and implementation of the system to meet both technical and user expectations.

5.2 Security Requirements

Security is the highest priority of the Bharat Ballot platform. Multiple layers of protection have been implemented to ensure the confidentiality, integrity, and authenticity of election data.

The platform shall:

- Authenticate all users before granting access to protected resources.
- Encrypt user passwords using bcrypt hashing.
- Use JWT-based authentication for secure session management.
- Enforce role-based access control (RBAC).
- Perform biometric face verification before allowing a citizen to vote.
- Prevent duplicate voting through one-person-one-vote enforcement.
- Restrict administrators to their assigned constituencies.
- Validate election status before permitting election operations.
- Protect all REST API endpoints from unauthorized access.
- Securely store voter and candidate images using cloud infrastructure.

5.3 Performance Requirements

The platform shall provide efficient response times to ensure a smooth user experience.

Performance expectations include:

- User authentication should complete within a few seconds under normal network conditions.

- Election dashboards should load efficiently.
- Candidate and voter searches should return results with minimal delay.
- Face verification should complete within an acceptable verification time.
- Vote submission should be processed immediately after successful verification.
- The system shall support simultaneous access by multiple authenticated users without noticeable performance degradation.

5.4 Reliability Requirements

The Bharat Ballot platform shall provide dependable and consistent operation throughout the election lifecycle.

The system shall:

- Maintain consistent election records.
- Preserve vote integrity.
- Prevent data corruption during transactions.
- Recover gracefully from temporary failures.
- Ensure that successfully submitted votes are permanently recorded.
- Maintain accurate election status information.

5.5 Availability Requirements

The platform shall remain available throughout active election periods to ensure uninterrupted election operations.

Availability requirements include:

- Continuous accessibility during election periods.
- Reliable cloud-hosted backend services.
- High availability of database services.
- Availability of the AI face verification service whenever biometric verification is required.
- Accessible web interfaces through supported browsers.

Scheduled maintenance should preferably occur outside active election periods.

5.6 Scalability Requirements

The Bharat Ballot platform has been designed with scalability in mind.

The system shall support:

- Multiple elections operating simultaneously.
- Increasing numbers of voters.
- Increasing numbers of candidates.
- Additional administrative users.
- Expansion to additional states and constituencies.
- Future deployment across larger geographical regions.

The modular architecture enables independent scaling of frontend applications, backend services, and AI components.

5.7 Usability Requirements

The platform shall provide a user-friendly experience for all categories of users.

The user interface shall:

- Provide clear navigation.
- Use consistent layouts across all portals.
- Present meaningful error messages.
- Support responsive design for different screen sizes.
- Minimize the number of steps required to complete election tasks.
- Display relevant information according to user roles.

Administrative interfaces shall simplify election management through organized dashboards and intuitive controls.

5.8 Maintainability Requirements

The Bharat Ballot platform shall be easy to maintain and extend.

Maintainability requirements include:

- Modular software architecture.
- Clearly separated frontend, backend, and AI services.
- Reusable software components.
- Organized project structure.

- Consistent coding standards.
- Comprehensive documentation.
- Easy integration of future modules.

The modular architecture allows individual services to be updated independently without affecting the entire platform.

5.9 Portability Requirements

The platform shall be deployable across different environments with minimal configuration changes.

The system supports:

- Cloud deployment.
- Local development environments.
- Cross-platform execution for backend services.
- Modern web browsers.
- Different operating systems capable of running Node.js and Python environments.

5.10 Compatibility Requirements

The Bharat Ballot platform shall remain compatible with commonly used technologies.

The platform supports:

- Google Chrome
- Microsoft Edge
- Mozilla Firefox
- Safari

Backend compatibility includes:

- Node.js Runtime
- MongoDB Atlas
- Python AI Service
- Cloudinary

5.11 Data Integrity Requirements

The system shall preserve the accuracy and consistency of election data throughout all operations.

The platform shall:

- Maintain unique voter identities.
- Prevent duplicate candidate registrations.
- Prevent duplicate votes.
- Preserve election history.
- Ensure secure vote recording.
- Validate relationships between elections, constituencies, candidates, and voters.
- Maintain database consistency during concurrent operations.

5.12 Privacy Requirements

The platform shall protect personal information collected during voter registration and election administration.

The system shall:

- Store sensitive information securely.
- Restrict access to personal records.
- Limit biometric information to authentication purposes.
- Prevent unauthorized disclosure of voter information.
- Protect authentication credentials from exposure.

5.13 Disaster Recovery Requirements

To minimize disruption during unexpected failures, the platform shall support recovery procedures.

Recovery considerations include:

- Database backup strategies.
- Secure storage of uploaded images.
- Recovery of election configuration data.
- Restoration of user accounts.

- Protection against accidental data loss.

5.14 Compliance Requirements

The Bharat Ballot platform has been designed with software engineering best practices and secure web application principles in mind.

The system adheres to:

- RESTful API design principles.
- Role-based access control.
- Secure password management.
- Modular software architecture.
- Cloud-native deployment practices.
- Standard web security guidelines.

Chapter Summary

This chapter defined the non-functional requirements that determine the quality, reliability, and operational characteristics of the Bharat Ballot platform. These requirements ensure that the system remains secure, scalable, reliable, maintainable, and user-friendly while supporting trustworthy digital election management. Together with the functional requirements presented in the previous chapter, these quality attributes establish the foundation for building a robust and production-ready election platform.

Chapter 6 — System Models and Diagrams

6.1 Introduction

System models provide a graphical representation of the Bharat Ballot platform and illustrate how different users, modules, and services interact throughout the election lifecycle. These diagrams improve understanding of the system architecture, business processes, data flow, and software components.

The following diagrams are included in this chapter:

- Use Case Diagram
- Activity Diagram
- Sequence Diagram
- Component Diagram
- Deployment Diagram
- Entity Relationship (ER) Diagram
- Data Flow Diagram (DFD)

These models complement the functional and non-functional requirements presented in the previous chapters and provide a visual representation of the overall system.

6.2 Use Case Diagram

Description

The Use Case Diagram illustrates the interactions between system users and the Bharat Ballot platform. It identifies the primary actors and the functionalities available to each user role.

Primary Actors

- Head
- Administrator
- Citizen (Voter)

Major Use Cases

Head

- Login
- Manage Elections
- Manage Administrators

- View Dashboard
- Publish Results
- Monitor Elections

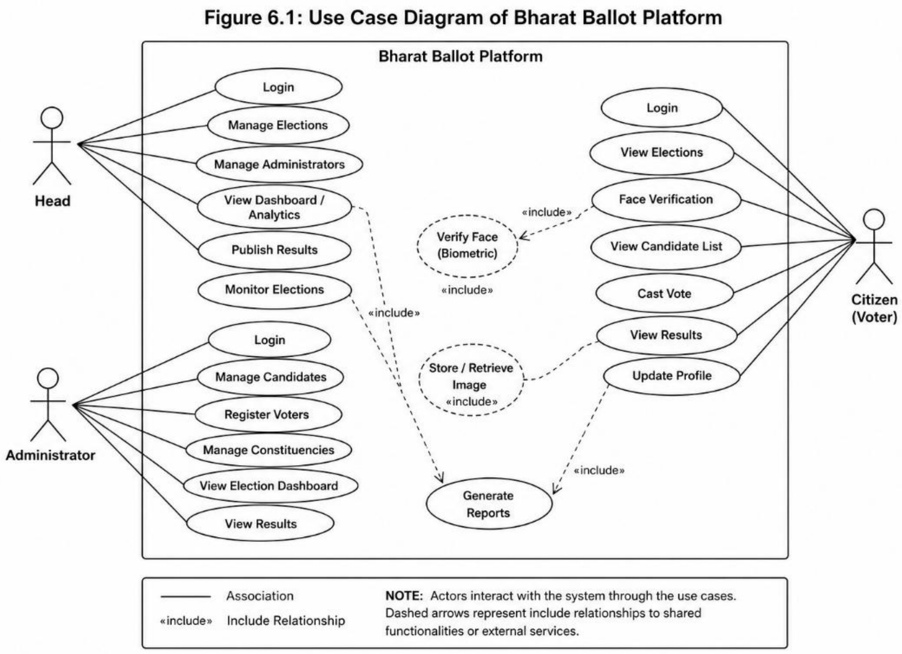
Administrator

- Login
- Manage Candidates
- Register Voters
- Manage Constituencies
- View Election Dashboard
- View Results

Citizen

- Login
- Face Verification
- View Elections
- Cast Vote
- View Results
- Update Profile

Figure 6.1: Use Case Diagram of the Bharat Ballot Platform



6.3 Activity Diagram

Description

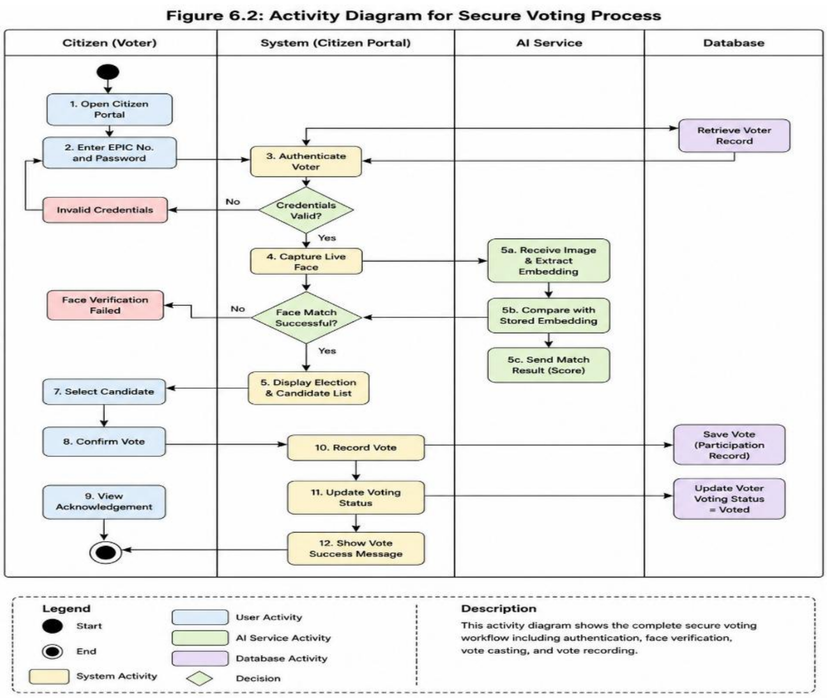
The Activity Diagram represents the sequence of activities performed during the voting process.

The process begins when the voter logs into the Citizen Portal and ends after a successful vote submission.

Workflow

1. Login
2. Authentication
3. Face Verification
4. Verification Success?
5. Display Candidate List
6. Cast Vote
7. Save Vote
8. Update Voting Status
9. Display Confirmation

Figure 6.2: Activity Diagram for Secure Voting



6.4 Sequence Diagram

Description

The Sequence Diagram illustrates the communication between the user, frontend application, backend server, AI service, and database during the voting process.

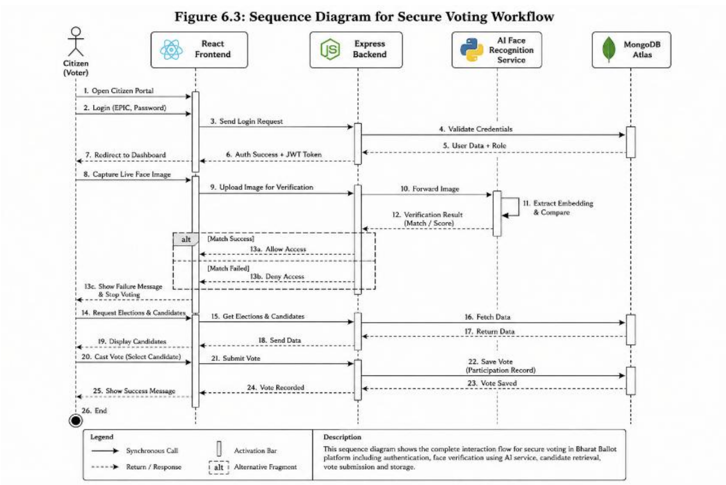
Participating Components

- Citizen
- React Frontend
- Express Backend
- AI Service
- MongoDB

Communication Sequence

1. Login Request
2. Authentication
3. Face Capture
4. Face Verification Request
5. AI Response
6. Candidate Request
7. Vote Submission
8. Vote Storage
9. Confirmation

Figure 6.3: Sequence Diagram for Voting Workflow



6.5 Component Diagram

Description

The Component Diagram illustrates the major software components of Bharat Ballot and their relationships.

Components

Frontend Applications

- Citizen Portal
- Command Center

Backend

- Express API Server

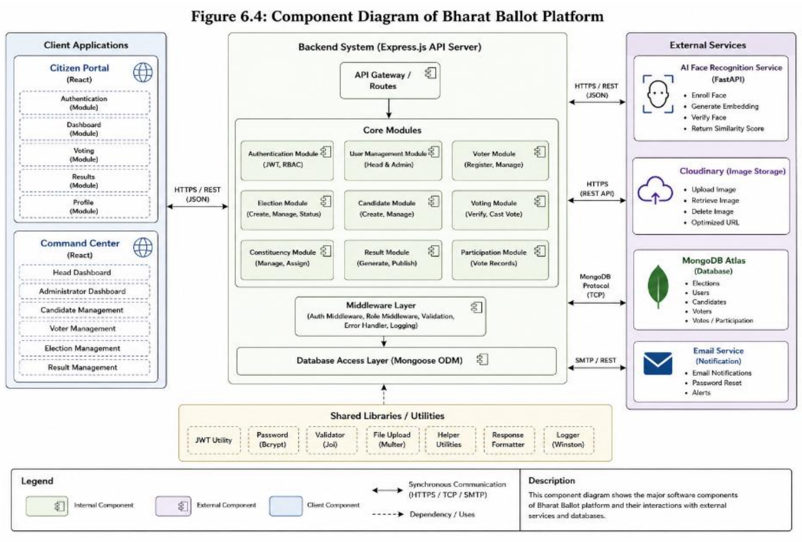
Services

- Authentication Module
- Candidate Module
- Voter Module
- Result Module

External Services

- MongoDB Atlas
- Cloudinary
- AI Face Recognition Service

Figure 6.4: Component Diagram



6.6 Deployment Diagram

Description

The Deployment Diagram illustrates how Bharat Ballot is deployed across cloud infrastructure.

Deployment Nodes

Client

- Browser

Cloud

- Render Frontend
- Render Backend

Database

- MongoDB Atlas

Cloud Storage

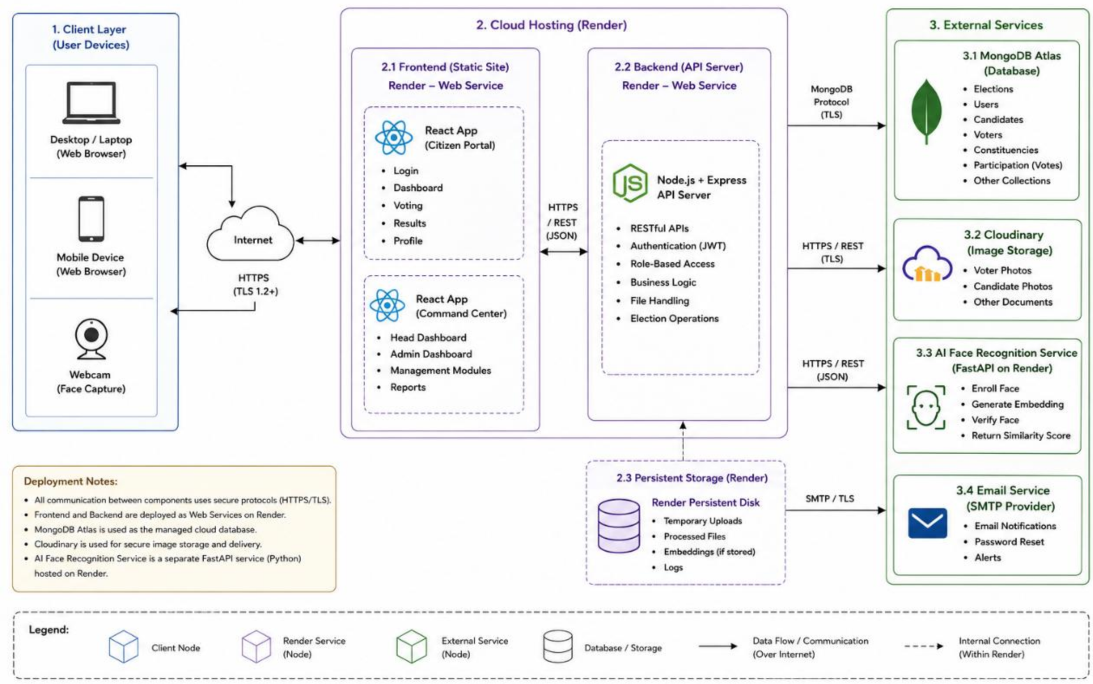
- Cloudinary

AI Server

- FastAPI Face Recognition Service

Figure 6.5: Deployment Diagram

Figure 6.5: Deployment Diagram of Bharat Ballot Platform



6.7 Entity Relationship (ER) Diagram

Description

The Entity Relationship Diagram illustrates the relationships between major database entities.

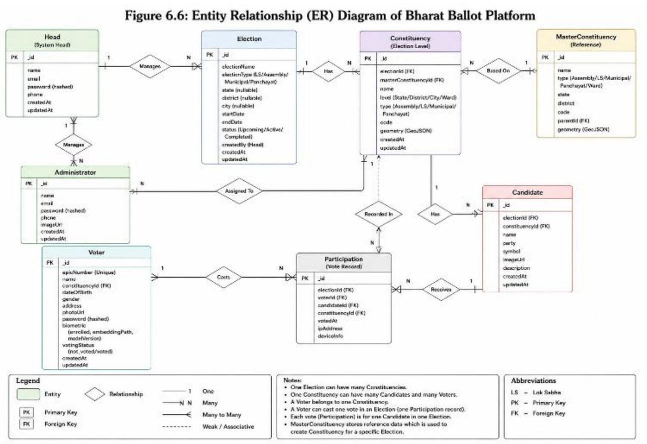
Primary Entities

- Head
- Administrator
- Election
- Candidate
- Voter
- Constituency
- Master Constituency
- Participation (Vote Record)

Key relationships include:

- One Election contains multiple Constituencies.
- One Constituency contains multiple Candidates.
- One Constituency contains multiple Voters.
- One Voter may participate in multiple elections over time, with each participation recorded separately.
- Administrators are assigned to one or more Constituencies.
- Candidates belong to both an Election and a Constituency.

Figure 6.6: Entity Relationship Diagram



6.8 Data Flow Diagram (DFD)

Level 0 DFD

The Level 0 Data Flow Diagram represents Bharat Ballot as a single system interacting with external users.

External Entities

- Head
- Administrator
- Citizen

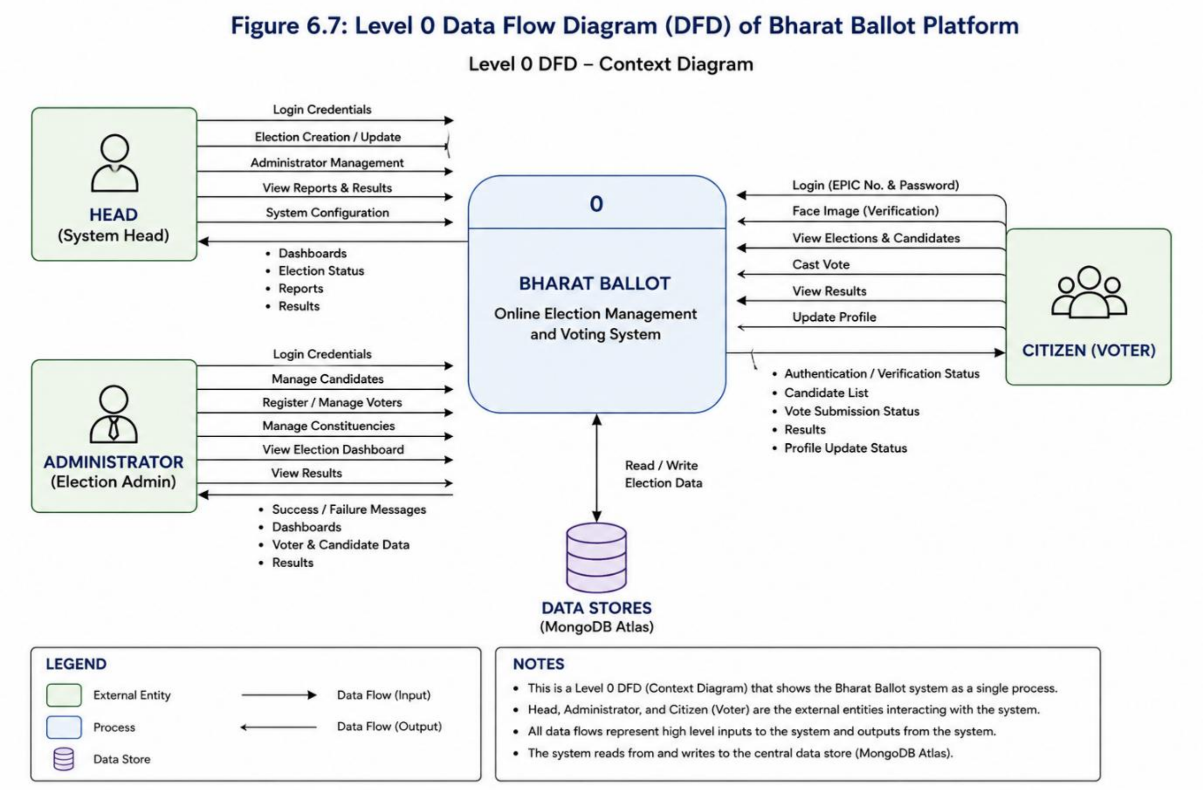
System

- Bharat Ballot

Outputs

- Election Information
- Voting Services
- Results

Figure 6.7: Level 0 Data Flow Diagram



Level 1 DFD

The Level 1 DFD decomposes the Bharat Ballot system into major functional processes.

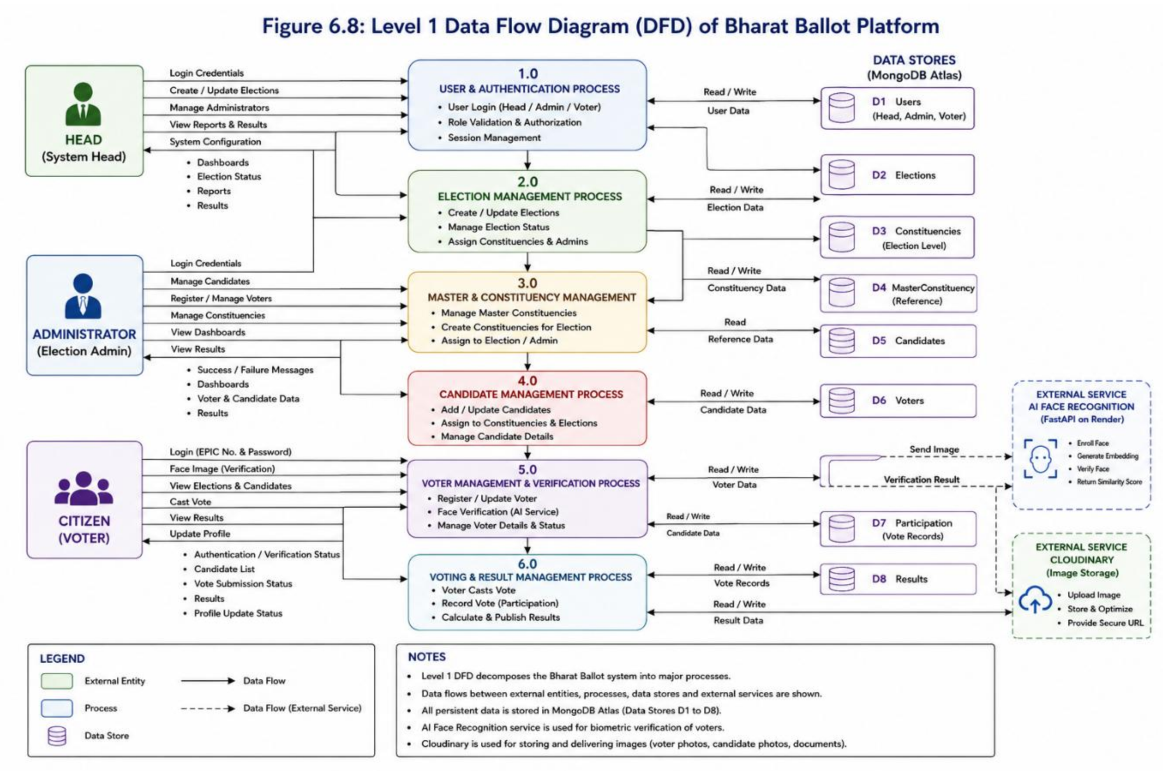
Processes

- Authentication
- Election Management
- Candidate Management
- Voter Management
- Face Verification
- Voting
- Result Processing

Data Stores

- Election Database
- Voter Database
- Candidate Database
- Participation Records

Figure 6.8: Level 1 Data Flow Diagram



Chapter Summary

This chapter presented the graphical system models used to describe the Bharat Ballot platform. Through UML diagrams, deployment architecture, ER modeling, and data flow representations, the chapter provides a comprehensive visual understanding of the system's structure, interactions, and operational workflow. These models complement the textual requirements specified in previous chapters and serve as valuable references for implementation, testing, maintenance, and future enhancements.

Chapter 7 – Future Scope

7.1 Introduction

Bharat Ballot has been developed as a secure and scalable digital election management platform capable of supporting multiple categories of elections. While the current implementation successfully manages election administration, voter registration, biometric verification, secure voting, and result publication, the modular architecture of the platform allows it to be extended with additional technologies and functionalities.

The following enhancements represent potential future developments that could further improve the security, transparency, accessibility, and efficiency of the platform.

7.2 Aadhaar e-KYC Integration

The platform can be integrated with government-approved digital identity verification systems such as Aadhaar e-KYC (subject to legal and regulatory approvals). This would enable:

- Automatic identity verification during voter registration.
- Reduction of duplicate voter records.
- Faster voter onboarding.
- Improved authenticity of voter information.

Such integration would reduce manual verification efforts while maintaining compliance with applicable government regulations.

7.3 Mobile Application

A dedicated Android and iOS application can be developed to increase accessibility for voters and election officials.

Potential features include:

- Secure voter login.
- Push notifications.
- Election reminders.
- Face verification.
- Mobile voting (where legally permitted).
- Live election updates.
- Result notifications.

The application could share the existing backend infrastructure through RESTful APIs.

7.4 Advanced AI-Based Fraud Detection

The current platform performs biometric face verification before voting. Future versions may incorporate additional artificial intelligence techniques to identify suspicious activities, such as:

- Multiple login attempts from different locations.
- Abnormal voting patterns.
- Repeated verification failures.
- Device anomaly detection.
- Automated fraud alerts for administrators.

These capabilities would strengthen election integrity through proactive monitoring.

7.5 Blockchain-Based Vote Audit Trail

Blockchain technology can be incorporated to create an immutable audit trail of election records.

Potential advantages include:

- Tamper-resistant vote records.
- Transparent audit mechanisms.
- Increased public trust.
- Secure historical election archives.
- Distributed verification of election integrity.

The blockchain layer would complement the existing database rather than replace it.

7.6 Observer and Monitoring Portal

A dedicated Election Observer Portal may be introduced for authorized observers and supervisory authorities.

Features may include:

- Real-time election monitoring.
- Live voter turnout statistics.
- Constituency-wise dashboards.

- Incident reporting.
- Administrative alerts.
- Election analytics.

This module would improve transparency and operational oversight.

7.7 Advanced Analytics Dashboard

Future versions may incorporate comprehensive analytical dashboards for election authorities.

Potential capabilities include:

- Voter turnout analysis.
- Constituency performance comparison.
- Candidate statistics.
- Historical election trends.
- Interactive charts and visualizations.
- Exportable reports.

These insights would support data-driven election management.

7.8 Multi-Language Support

To improve accessibility across diverse regions, the platform may support multiple Indian languages.

Potential languages include:

- English
- Hindi
- Punjabi
- Tamil
- Telugu
- Bengali
- Marathi
- Gujarati
- Kannada

- Malayalam

A multilingual interface would improve usability for voters and election officials from different linguistic backgrounds.

7.9 Geographic Information System (GIS) Integration

Future versions of Bharat Ballot can integrate GIS technologies for spatial visualization of election data.

Possible applications include:

- Interactive constituency maps.
- Polling station visualization.
- Administrative jurisdiction mapping.
- Geographic election analytics.
- Region-wise voter statistics.

GIS integration would assist election administrators in planning and monitoring electoral activities.

7.10 Cloud-Native Scaling

As election participation increases, the platform can be enhanced using cloud-native technologies.

Future improvements may include:

- Containerization using Docker.
- Kubernetes orchestration.
- Auto-scaling backend services.
- Distributed caching.
- Load balancing.
- High-availability deployment.

These enhancements would improve performance during large-scale elections.

7.11 Notification and Communication System

A centralized notification service can improve communication between election authorities and users.

Potential features include:

- Email notifications.
- SMS alerts.
- Election reminders.
- Password reset notifications.
- Result announcements.
- Administrative broadcasts.

This module would enhance user engagement and operational efficiency.

7.12 Accessibility Enhancements

To make Bharat Ballot more inclusive, future versions can incorporate additional accessibility features.

Possible improvements include:

- Screen reader compatibility.
- High-contrast display modes.
- Adjustable text sizes.
- Keyboard navigation.
- Voice-assisted navigation.
- Accessibility compliance based on WCAG guidelines.

These features would ensure broader usability for users with different accessibility needs.

7.13 National-Level Deployment

The current implementation demonstrates the feasibility of secure digital election management. With appropriate legal, security, and infrastructural enhancements, the platform can be extended to support elections at larger administrative scales.

Potential deployment targets include:

- Universities and educational institutions.
- Cooperative societies.
- Corporate elections.
- Local government bodies.

- Municipal corporations.
- State-level elections.
- National-level election management (subject to regulatory approval).

The modular architecture of Bharat Ballot provides a foundation for such future expansion.

Chapter Summary

This chapter outlined the potential future enhancements of the Bharat Ballot platform. These proposed developments leverage emerging technologies such as artificial intelligence, blockchain, cloud computing, GIS, and mobile applications to further improve election security, transparency, scalability, and accessibility. The platform's modular architecture ensures that these enhancements can be incorporated progressively without significant redesign, allowing Bharat Ballot to evolve alongside future technological and organizational requirements.